EEG PROJECT

NEURAL CORRELATES OF FEEDBACK PROCESSING IN COMPETITIVE DECISION-MAKING USING EEG



Indian Institute Of Technology, Gandhinagar

Course: Human brain mapping

Submitted to

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Abstract

The current study explored the brain basis of feedback processing with the help of an ERP analysis in the context of the competitive rock-paper-scissors game (N=20). More specifically, the study investigated neural responses associated with the feedback-related negativity (FRN) and the P300 components across different feedback conditions (WIN, LOSS, DRAW). Although grand-average ERP waveforms suggested greater frontocentral negativity for LOSS outcomes relative to WIN outcomes during the FRN interval, these differences did not reach statistical significance, suggesting a possible trend toward enhanced performance-monitoring activity following unfavorable feedback. There were no significant differences in P300 amplitude between the win and loss conditions; however, an exploratory analysis revealed distinct modulations of the P300 component in response to DRAW condition feedback. Finally, a non-significant but suggestive trend was observed for the effect of participants' performance on feedback processing.

1. Introduction

Adaptive decision-making in the context of competition requires continuous assessment of feedback to enable behavioral change and enhance future decision-making skills. Games that involve competitive actions like RPS necessitate rapid assessment of win, lose, and uncertain outcomes while continuously adjusting action based on performance.

1.1Background and rationale

Adaptive decision-making in such scenarios results from brain functions that promote reinforcement learning and performance monitoring (Holroyd & Coles, 2002). The use of an

electroencephalogram (EEG) combined with event-related potentials (ERPs) can help analyze these processes, as they have very high temporal resolution.

1.2 Review of relevant literature

Some of the ERP components most frequently used in studies of feedback processing include the Feedback-Related Negativity (FRN) and the P300. Specifically, the FRN component is characterized by a frontocentral negative deflection occurring around 250–350 ms after feedback onset, and it is strongly linked to unfavorable outcomes, reward prediction errors, and performance monitoring (Holroyd & Coles, 2002). At the same time, the P300 is known to occur around 300-500 ms after feedback and is associated with attention allocation, salient stimuli, and the evaluation of behavioral relevance.

Similarly, competitive decision-making processes can be characterized by uncertainty and ambiguity, especially during draws when neither reward nor punishment is clearly defined. Unlike clear-cut win-loss situations, draw conditions are ambiguous and require further cognitive evaluation and interpretation. Thus, feedback evaluation involving ambiguity can engage different brain processes in their subsequent phases of processing.

This current study explored brain activities involved in feedback evaluation during a competitive Rock-Paper-Scissors game using EEG data from the open-source hyperscanning dataset of Moerel et al. (2025). The analysis was based on feedback-locked ERPs and focused on brain activity in the WIN, LOSS, and DRAW conditions, particularly the FRN and P300 responses. Along with the main ERPs investigation, an exploratory analysis was conducted on draw conditions and individual competitive performance.

1.3 Research Questions and Hypotheses

Objective

The primary purpose of this research is to investigate the brain processes underlying the discrimination of competitive feedback during decision-making. Specifically, the WIN, LOSS, and DRAW outcomes of the Rock-Paper-Scissors competition were evaluated.

Research questions included the following:

- Are the LOSS results characterized by higher FRN activity than the WIN results when evaluating competitive feedback?
- Are the WIN results characterized by higher P300 activity than the LOSS results?
- Do DRAW outcomes evoke distinct ERP responses compared to WIN and LOSS outcomes?
- Is there any difference between the neural response patterns of high-performers and other participants?

Hypotheses

- H1: LOSS outcomes will elicit greater FRN negativity than WIN outcomes.
- H2: WIN outcomes will produce larger P300 amplitudes than LOSS outcomes.
- H3: DRAW outcomes will evoke distinct ERP responses, especially during later processing stages.
- H4: Higher-performing participants will show different neural responses to feedback compared with lower-performing participants.

2. Methods

2.1 Dataset, Participants, and Task Protocol

For the present study, the EEG data from the Rock-Paper-Scissors competitive hyperscanning dataset were used. This dataset consists of simultaneous EEG recordings collected during competitive Rock-Paper-Scissors (RPS) games. During the task, participants selected responses and received feedback indicating whether they had won, lost, or drawn.

In this current research, a sample size of 22 participants was selected at first, but later, one participant had no EEG data files, and another participant did not have enough usable epochs post-preprocessing, thus leaving a total of 20 participants for further analysis.

2.2 Neurophysiological data acquisition

EEG data were recorded with the use of the BioSemi recording equipment with a sampling rate of 2048 Hz. EEG recordings were acquired simultaneously from interacting participants using a hyperscanning paradigm designed to capture neural activity during real-time competitive decision-making. For each participant, 64 EEG channels were available, along with behavioral events that contained information about the trials and their outcomes. All EEG data processing, ERP calculation, and statistical analyses were done through the MNE-Python library (Gramfort et al., 2013).

The original hyperscanning dataset contained simultaneous EEG recordings from two interacting participants within each recording session. During preprocessing, channels specific to Player 1 were isolated for analysis to avoid cross-participant mixing of neural activity. Initial analyses used a broader electrode-selection approach before the detailed examination of the dataset structure. After revisiting the dataset documentation and associated publication, a theory-driven

region-of-interest approach focusing on frontocentral electrodes relevant to feedback processing was implemented for the final analyses.

2.3 EEG Preprocessing

The preprocessing procedures for the EEG were performed using the open-source MNE-Python toolbox (Gramfort et al., 2013) in Python. First, for each participant, the electrodes corresponding to player one were identified in the acquired data during the process of hyperscanning. These electrodes were renamed according to the BioSemi64 spatial montage system. The spatial positioning of these electrodes was also identified based on the standard montage.

To reduce computational costs without compromising the temporal resolution required for ERP analysis, the initial sampling rate of 2048 Hz was downsampled to 256 Hz. Also, bandpass filtering was applied over the frequency range 0.1-30 Hz.

Behavioral files describing the trials and outcome were used to extract feedback-locked events for ERP analysis. Since feedback was presented about 4 seconds after trial onset in the experimental setting, the time stamps were changed correspondingly prior to epoch extraction. The period between –200 ms and 800 ms from the feedback onset was considered an epoch of interest.

The epochs were baseline-corrected using the pre-feedback period between –200 ms and 0 ms. Automatic artifact removal was performed, excluding epochs with amplitudes greater than ±250 μV.

2.4 ERP Analysis

ERP analysis examined neural responses to WIN, LOSS, and DRAW feedback. In each condition, feedback-locked ERPs were calculated by averaging ERPs over the course of all trials.

The primary area of interest was the frontocentral region, including electrodes Fz, FC1, FC2, FCz, Cz, C1, and C2, because these electrodes are highly relevant to feedback monitoring and the Feedback-Related Negativity (FRN).

Two ERP components were measured in the present experiment. The first component, the Feedback-Related Negativity (FRN), was measured in terms of the mean amplitude of this ERP component between 250 and 350 ms after the presentation of feedback, which represents the time window when the unfavorable outcome is processed along with reward prediction error processing. The second ERP component, the P300, was measured based on the mean amplitude between 300 and 500 ms.

Mean ERP amplitudes of each component were determined by averaging data from the electrodes in the defined frontocentral region. Grand averages of ERP waveforms were then obtained based on the differences in WIN, LOSS, and DRAW outcomes.

2.5 Statistical Analysis

Statistical analyses were carried out to compare ERP amplitudes across feedback types (WIN, LOSS, and DRAW). In order to determine differences within subjects regarding FRN and P300 amplitudes under different feedback conditions, paired samples t-tests were applied. In terms of primary analyses, comparisons between ERP amplitudes of WIN versus LOSS conditions were made for the assessment of differential neural processing related to advantageous versus disadvantageous competitive feedback.

Besides, two exploratory analyses were performed. The first one involved comparing ERP patterns associated with ambiguous feedback by examining ERPs in DRAW conditions to detect

neural specificity. The other exploratory analysis examined individual differences in feedback processing as a function of competition outcomes.

For the purpose of competitive performance assessment, participants' behavior-based scores reflecting individual competitive performance levels were calculated as the difference between the number of wins and losses across the task. Participants were divided into two groups based on their competitive performance scores. An independent-samples t-test was then conducted to compare ERP amplitudes between the two groups.

3. Results

3.1 Participant Quality Control

Following EEG pre-processing and artifact rejection, one participant was excluded due to missing EEG files, and another was excluded after preprocessing due to insufficient usable epochs following artifact rejection.

The remaining 20 participants had sufficient usable epochs for the conditions WIN, LOSS, and DRAW, allowing ERPs to be computed for each condition separately. The visual check of the pre-processed EEG files and grand-averaged waveforms confirmed that the artifacts were successfully minimized.

3.2 Grand Average ERP Responses

Grand-average ERPs were calculated among all subjects for three conditions (WIN, LOSS, and DRAW) in terms of the chosen frontocentral region of interest (Fz, FC1, FC2, FCz, Cz, C1, and C2). The obtained ERPs are shown in Figure 1 below. It was observed that grand-average waveforms showed individual brain potentials depending on outcome type. Within the FRN time window (250–350 ms), grand-average ERP waveforms suggested greater frontocentral negativity for LOSS outcomes relative to WIN outcomes, which may reflect enhanced neural processing of

negative feedback and performance monitoring.DRAW events showed an intermediate level of brain activity during this period.

In the latter P300 period (300-500 ms), DRAW events showed distinct waveform patterns compared to WIN and LOSS conditions. It can be argued that ambiguous results require greater levels of attention and evaluation compared to clearly positive or negative results. In other words, ERP amplitudes appeared greater for ambiguous feedback during later stages of processing.

Topographic scalp maps for FRN and P300 periods are depicted in Figure 2. There was stronger frontocentral negativity after LOSS events in the former period, while DRAW events had more distinctive patterns in the latter period.

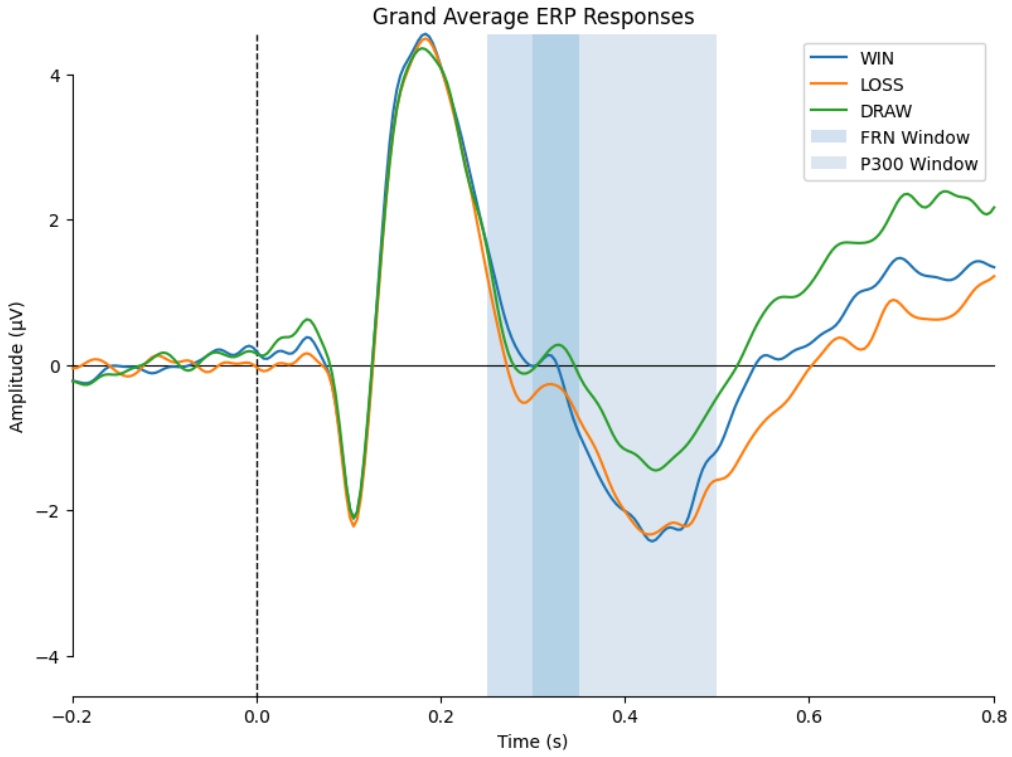


Figure 1. Grand-average ERP waveforms for the WIN, LOSS, and DRAW conditions measured across the frontocentral electrode sites (Fz, FC1, FC2, FCz, Cz, C1, and C2). The ERPs were locked to feedback onset, and baseline correction was applied from -200 ms to 0 ms.

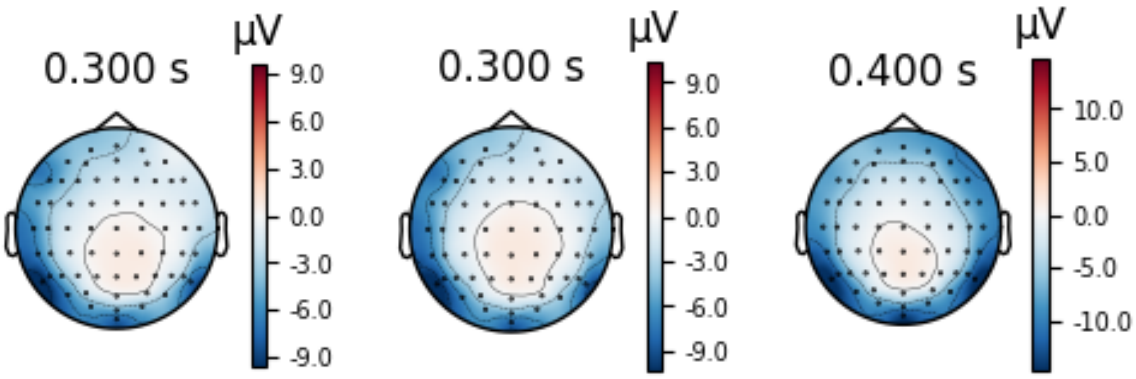


Figure 2. Topographical scalp distributions of ERP activity during competitive feedback processing.

In fig 2, the left panel shows the LOSS condition during the Feedback-Related Negativity (FRN) interval at approximately 300 ms, the middle panel shows the WIN condition during the FRN interval at approximately 300 ms, and the right panel shows the DRAW condition during the P300 interval at approximately 400 ms. Greater frontocentral negativity was observed for the LOSS condition relative to the WIN condition during the FRN interval, whereas the DRAW condition demonstrated distinct later-stage activity during the P300 interval, suggesting differential neural processing of ambiguous competitive feedback.

3.3 FRN Results

Mean FRN amplitudes for WIN, LOSS, and DRAW conditions are presented in Figure 3. Although grand-average ERP waveforms suggested greater frontocentral negativity for LOSS

outcomes relative to WIN outcomes, paired-samples t-tests did not reveal a statistically significant difference between LOSS and WIN conditions, t(19) = -1.446, p = .165.

Exploratory analyses involving DRAW outcomes similarly did not reveal significant FRN differences between DRAW and WIN conditions, t(19) = 0.226, p = .824, or between DRAW and LOSS conditions, t(19) = 1.507, p = .148.

These findings suggest that although visual ERP trends indicated enhanced negativity following unfavorable feedback, FRN differences across competitive outcomes were not statistically reliable in the present sample.

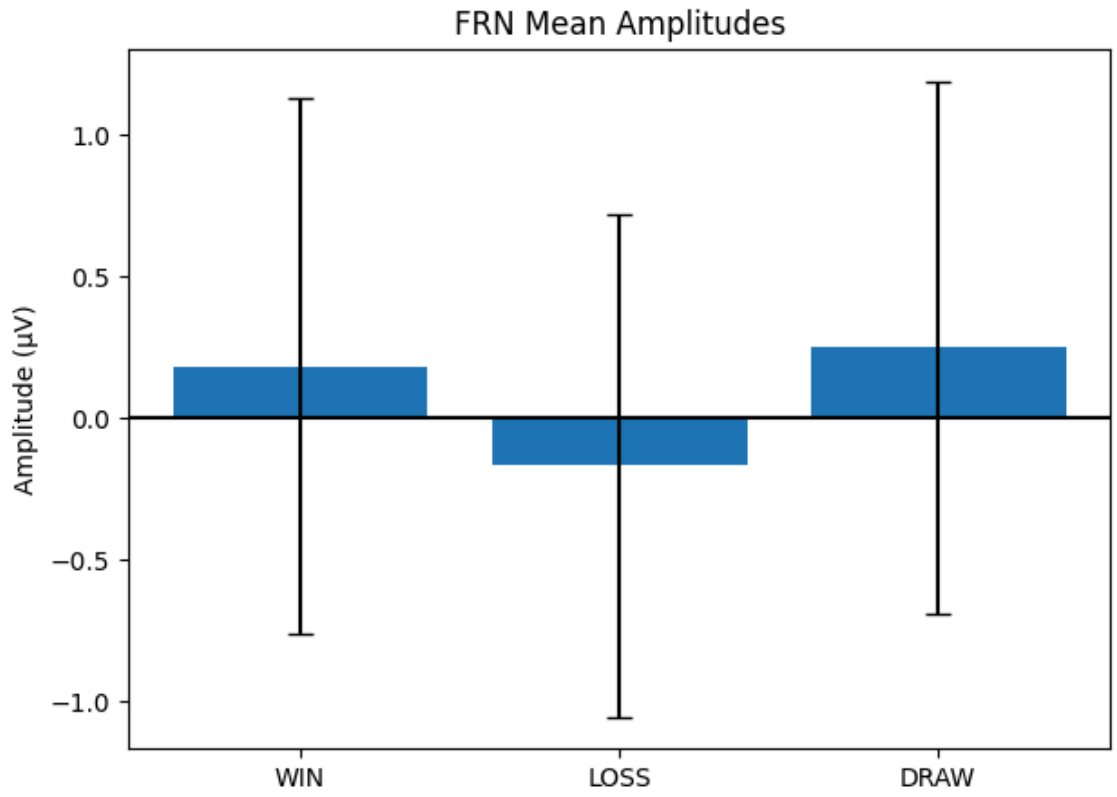


Figure 3. *Mean FRN amplitudes for WIN, LOSS, and DRAW conditions computed across the frontocentral region of interest. Error bars represent the standard error of the mean (SEM) across participants. Although LOSS outcomes showed a trend toward greater frontocentral*

negativity relative to WIN outcomes, differences between conditions were not statistically significant.

3.4 P300 Results

The mean P300 amplitudes in relation to the WIN, LOSS, and DRAW conditions are shown in Figure 4. There was no statistical difference in P300 amplitudes between the WIN and LOSS outcomes, $t(19) = -0.203$, $p = .842$, suggesting that late-stage attention and evaluation are equalized across both positive and negative competitive outcomes.

Nonetheless, there were statistically significant differences when examining the relationship between P300 amplitudes and DRAW conditions compared to WIN and LOSS outcomes, respectively. The results showed that P300 amplitudes for DRAW outcomes were significantly different from those for WIN outcomes, $t(19) = 2.363$, $p = .029$, and for LOSS outcomes, $t(19) = 2.382$, $p = .028$.

Thus, the present experiment demonstrates the involvement of specific neural mechanisms during the late stage of feedback processing in response to ambiguous competitive outcomes. Indeed, an inspection of the ERP waveforms showed the increased positive deflection related to the ambiguous outcome conditions during the P300 time window.

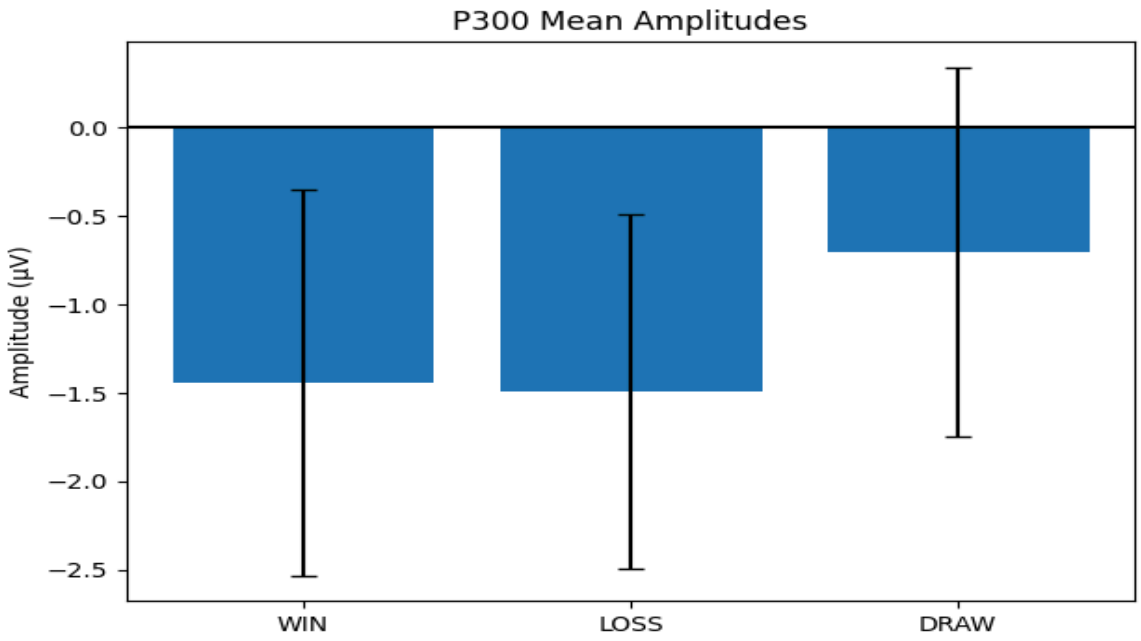


Figure 4. The average amplitudes of the P300 ERP component for WIN, LOSS, and DRAW conditions were calculated from the FC ROI. Error bars denote SEMs. Although there was no statistically significant difference between WIN and LOSS conditions, DRAW conditions exhibited significantly different amplitudes of the P300 ERP component.

3.5 Exploratory Winner versus Loser Analysis

To explore inter-individual differences in feedback processing, subjects were divided into two groups based on their performance, as measured by the difference between total wins and total losses during the experiment. Independent-samples t-tests were conducted to compare ERP activity between higher and lower performers.

The results showed a non-significant trend toward greater FRN negativity in higher performers than in lower performers during LOSS trials, $t(18) = -1.602$, $p = .127$. Even though this result failed to reach statistical significance, it suggests that higher performers may show a stronger neural response to loss events.

It is possible that higher performers may have better performance-monitoring or feedback-processing abilities in competitive situations. However, since the above test was conducted on an exploratory basis and with a relatively small sample size, the result should be taken with caution.

4. Discussion

The present study investigated neural correlates of feedback processing during competitive decision-making using EEG-based ERP analysis. Although grand-average ERP waveforms suggested enhanced frontocentral negativity following LOSS outcomes relative to WIN outcomes during the FRN interval, statistical analyses did not reveal significant differences between conditions.

Nevertheless, the observed waveform patterns were broadly consistent with the Reinforcement Learning Theory of the FRN (Holroyd & Coles, 2002), according to which the FRN represents a disutility signal that emerges from a negative discrepancy between actual and expected outcomes. In competitive situations, negative outcomes play a crucial role as Reward Prediction Errors (Walsh & Anderson, 2012), indicating that a new behavioral strategy is required. Therefore, the observed waveform trends may reflect neural mechanisms associated with performance monitoring and behavioral adaptation during competitive feedback processing.

However, the primary hypothesis that P300 amplitudes would be greater for WIN than for LOSS outcomes was not supported. In contrast to other money-gambling paradigms, where winning events elicit larger P300 components due to their greater motivational relevance (Yeung & Sanfey, 2004), in Rock-Paper-Scissors, losing events seem to have the same level of

"information value" for future choices as winning events do. The lack of discrimination supports the Context-Updating Theory by Donchin & Coles (1988), which posits that P300 can be used in this paradigm to indicate the informational significance of feedback rather than its positive/negative valence alone.

Importantly, exploratory analyses revealed significant differences in P300 amplitudes between DRAW outcomes and both WIN and LOSS conditions. Although no significant FRN differences were observed between DRAW outcomes and the other conditions, differences in P300 activity suggest that ambiguous feedback in competitive contexts may engage additional attentional resources during later stages of processing. It has been established that ambiguous feedback increases the cognitive effort required to assimilate it into internal model representations of the environment (Huettel et al., 2006).

Individual Differences and Strategy

This emerging exploration of increased FRN negativity for those performing better in the LOSS condition is important. Although the exploratory winner-versus-loser analysis did not reach statistical significance, a trend toward greater FRN negativity among higher-performing participants was observed during LOSS trials. This pattern may suggest increased neural sensitivity to unfavorable feedback among more successful competitors, potentially facilitating enhanced error monitoring and adaptive strategy modification. Similar interpretations have previously been proposed in ERP studies examining the relationship between FRN amplitude and feedback-guided behavioral adaptation (San Martín, 2012).

5. Limitations and Future Directions

A few limitations must be considered when interpreting the results of the present study.

1. A relatively small number of participants was involved in the current study, potentially diminishing statistical power, particularly in individual-difference research.
2. While EEG recordings offer superb temporal resolution, their spatial resolution is quite poor, making it difficult for researchers to identify the brain generators responsible for the effect.
3. The current study primarily focused on frontocentral ERP components, leaving aside more sophisticated approaches such as source localization or functional connectivity analysis.

Future research in the field might focus on enhancing the current results using different methodologies, such as source-localization or connectivity-based analysis techniques, larger participant groups, and the use of competitive feedback information in a dynamic context within multiple trials.

6. Conclusion

The present experiment was designed to study the effect of feedback responses to competitive outcomes in the Rock-Paper-Scissors game using ERP analyses based on EEG data. Grand-average ERP waveforms suggested enhanced frontocentral FRN responses following LOSS outcomes relative to WIN outcomes; however, these differences did not reach statistical significance. Despite no significant difference in P300 amplitude between WIN and LOSS conditions, the results show the distinctive P300 response to DRAW outcomes. This suggests that ambiguous competition may trigger additional attention and evaluation mechanisms during late feedback processing. Additionally, exploratory comparisons between winners and losers

revealed an emerging difference in loss feedback responses among high- and low-performers. This work adds to the knowledge on the neural basis of competitive behavior and illustrates the utility of ERP techniques in studying feedback cognitive processes in social contexts. Moreover, this study emphasizes the significance of incorporating ambiguous competition results, namely DRAW, in studying the dynamic process of feedback perception in humans.

References

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- Dataset link-https://openneuro.org/datasets/ds006761

Many of the points mentioned during the presentation have changed after further analysis of the data. The paper attached to the dataset we had not found earlier, so we followed the mid-variance approach during the presentation. But now we have done it through proper montage mapping, as per the paper.